Inter-annotator Robustness and Structural Primacy of Gradients in a Gradient-Based Organizational Classification

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Abstract

A gradient-based classification of heterogeneous complex systems was proposed in a companion paper, in which organizational regime boundaries are defined by inter-axis gradients ($\Delta_{23} = A_2 - A_3$, $\Delta_{45} = A_4 - A_5$) rather than by absolute axis values. The present study tests whether this gradient structure is robust across independent annotators. Twenty-nine systems spanning biological, technological, institutional, economic, and social domains were scored by three annotators: a human–LLM pair (A0), and two independent LLMs (Gemini, GPT-4), each working from dedicated textual sources. Results show that the normative–revision gradient Δ_{45} is robustly preserved across all annotator pairs (Spearman $ρ = 0.69$–0.88, all $p < 0.001$), while the propagation–integration gradient Δ_{23} is robust between A0 and Gemini ($\rho = 0.78$) but sensitive to the calibration of the integration axis when scored by GPT-4 ($\rho = 0.58$). The inter-system distance matrices are correlated at $ρ = 0.63$–0.76 across annotator pairs, indicating that the relational structure of the organizational space is preserved even when absolute scores diverge. A reinforced prompt protocol corrects identified framing errors (historical vs.\ operational scoring, endogenous vs.\ imposed normativity) and reduces severe divergences (≥ 0.30) from 14.5\

1 Introduction

A companion paper [1] proposed to classify heterogeneous complex systems using five axes—hierarchical depth (A_1), propagation capacity (A_2), integration (A_3), normativity (A_4), and revision capacity (A_5)—and showed that the gradients between axes, particularly $\Delta_{23} = A_2 - A_3$ and $\Delta_{45} = A_4 - A_5$, are better discriminants of organizational regime than absolute axis values. A second paper [2] extended this analysis to 28 systems across six domains and found low effective dimensionality (PCA participation ratio = 3.05), stable gradient ranks under scoring perturbation, and four emergent morphodynamic families.

However, all prior results relied on a single annotator configuration. This raises an immediate question: is the structure found in [2] a property of the systems, or a property of the scorer?

The present paper addresses this question directly. Its central thesis is:

> What the morphodynamic index captures is not a set of absolute positions, but a pattern of relational orderings between systems—proximities, separations, and gradient ranks—that is partially invariant under change of annotator. The gradients function

as relational invariants under scoring perturbation, not as intrinsic coordinates in an absolute sense.

This is a weaker claim than "the gradients measure real properties of systems" and a stronger claim than "the gradients are artefacts of the protocol." It asserts that the instrument produces a structurally consistent relational geometry, without claiming that this geometry is fundamental. The distinction matters: a relational invariant can be useful as a coordinate system even if the underlying axes are imperfectly calibrated, provided the relations between measurements are preserved.

To test this thesis, we scored 29 systems with three independent annotator configurations and evaluated four hypotheses:

- **H1 (Convergence):** Mean absolute inter-annotator difference ≤ 0.15 on all axes.
- **H2 (Gradient robustness):** Spearman $\rho > 0.70$ for gradient ranks (Δ_{23}, Δ_{45}) between annotator pairs.
- **H3 (A3–A4 constraint):** The correlation between integration and normativity persists across annotators.
- **H4 (Gradient primacy):** Inter-annotator variance on gradients is lower than on individual axes.

2 Method

2.1 Systems

Twenty-nine systems were scored, spanning six domains: biological (7 systems: eukaryotic cell, ant colony, gut microbiome, coral reef, temperate forest, biological neural network, human psyche), technological (6: Linux kernel, static LLM, Internet BGP, electrical grid, PostgreSQL database, domestic thermostat), institutional (7: ECB, Benedictine order, US Constitution, bureaucracy, institutional archive, Prussian army, parliament in crisis), economic (4: global supply chain, local market, Bitcoin blockchain, industrial pipeline), social (3: Wikipedia, Occupy Wall Street, artist collective), and scientific (2: NaCl crystal, startup in pivot).

2.2 Annotators

Three annotator configurations were used:

1. **A0 (reference):** Human scorer assisted by Claude (Anthropic), scoring in dedicated conversation threads. Each system was scored independently using 3 textual sources per system. This is the reference configuration from [2].
2. **Gemini:** Google Gemini, scoring from the same or equivalent textual sources (provided as .epub, .pdf, or .txt files). Two rounds were performed: a first pass (v1) with a standard prompt, and a second pass (v2) with reinforced prompts for 12 systems identified as divergent.

3. **GPT-4:** OpenAI GPT-4, scoring from textual sources provided as .txt files. Standard prompt only (no reinforced pass).

All annotators used the same 25 sub-criteria protocol (5 per axis, scored 0/0.5/1). No annotator had access to the scores of the others.

2.3 Reinforced prompt protocol

The first Gemini pass revealed three systematic error types:

1. **Historical/operational confusion (A5):** The annotator scores revision capacity over the system's entire history rather than during active functioning. Affected: Prussian army (A5: 1.00 → 0.40), Constitution (A5: 1.00 → 0.60), archive (A5: 1.00 → 0.30).

2. **Endogenous/imposed confusion (A4):** The annotator attributes normativity to design constraints rather than system-produced norms. Affected: PostgreSQL (A4: 0.90 → 0.70).

3. **Structure/functioning confusion (A3, A4):** The annotator scores formal institutions rather than their effective functioning at the time evaluated. Affected: Parliament in crisis (A3: 1.00 → 0.20, A4: 1.00 → 0.10).

For each error type, a CRITICAL INSTRUCTIONS block was added to the scoring prompt, specifying the distinction to enforce and providing concrete examples. The full prompt protocol is available in the repository.

2.4 Statistical analysis

Inter-annotator agreement was assessed by mean absolute difference, proportion of severe divergences (≥ 0.30), and Spearman rank correlation on axis scores and gradient values. Structural preservation was assessed by correlating pairwise distance matrices (Euclidean distance on standardized axes) between annotators. The A3–A4 constraint was tested by Spearman correlation within each annotator's scores.

3 Results

3.1 Axis-level agreement (H1)

Table 1: Inter-annotator agreement on axis scores (25 systems with complete data for all 3 annotators).

Pair	Mean $	{Δ}	$	Median	Diffs ≥ 0.30
A0–Gemini v2	0.105	0.10	5.6\A0–GPT-4	0.112	0.10
11.2\Gemini–GPT-4	0.126	0.10	14.4\		

H1 is supported for A0–Gemini (mean $|\Delta| = 0.105 < 0.15$) and for A0–GPT-4 (0.112), but marginal for Gemini–GPT-4 (0.126). All annotators show a systematic bias pattern: A0 scores lower than the mean on all axes (mean bias −0.05), GPT-4 scores higher (mean bias +0.03), and Gemini v2 is approximately neutral.

The bias is not uniform across axes. GPT-4's positive bias is strongest on A_3 (integration, +0.03) and A_5 (revision, +0.04), which are precisely the axes that define the two main gradients. This axis-dependent bias has consequences for gradient robustness (see below).

3.2 Effect of reinforced prompts

Table 2: Effect of reinforced prompts on Gemini scoring (12 re-scored systems).

Metric	Gemini v1	Gemini v2
Mean $	{Δ}	$ vs A0
0.130	0.114	
Diffs ≥ 0.30	14.5\Max $	{Δ}
$	0.80	0.70
$\rho(\Delta_{23})$ vs A0	0.565	0.757
$\rho(\Delta_{45})$ vs A0	0.338	0.780

The reinforced prompts reduce severe divergences by 52\

Three error types were identified and corrected:

- The historical/operational error on A_5 was fully corrected in all affected systems.
- The endogenous/imposed error on A_4 was corrected in PostgreSQL and pipeline.
- The structure/functioning error on A_3/A_4 was corrected in the parliament in crisis.

Residual divergences after prompt reinforcement are interpretive rather than procedural: they reflect genuine disagreements about sub-criterion thresholds (e.g., whether apoptosis counts as resistance failure or as a regulatory mechanism for N5 in the eukaryotic cell).

3.3 Gradient robustness (H2)

Table 3: Spearman rank correlations on gradient values (25 complete systems).

Gradient	A0–Gemini	A0–GPT-4	Gemini–GPT-4	H2 threshold
Δ_{45}	**0.881**	**0.741**	0.685	0.70
Δ_{23}	**0.784**	0.582	0.495	0.70

Δ_{45} passes H2 for A0–Gemini and A0–GPT-4. Δ_{23} passes H2 only for A0–Gemini. The normative–revision gradient is the most robust inter-annotator coordinate of the instrument.

The fragility of Δ_{23} with GPT-4 is traceable to GPT-4's inflation of A_3 (integration) on systems where integration is partial or contested: global supply chain (A_3: 0.40 vs 1.00), Occupy Wall Street (A_3: 0.40 vs 0.80), startup (A_3: 0.50 vs 0.70). GPT-4 tends to score theoretical integration capacity rather than effective integration, a calibration issue that could be addressed by providing anchor examples in the prompt.

3.4 A3–A4 constraint (H3)

Table 4: Spearman correlation between A_3 and A_4 within each annotator.

Annotator	$\rho(A_3, A_4)$	\$p\$-value
A0	0.712	0.0001
Gemini v2	0.628	0.0008
GPT-4	0.422	0.035

The correlation persists across all three annotators (all $p < 0.05$). It is strongest for A0, weaker for GPT-4, but always positive and significant. H3 is supported: the A3–A4 coupling is not an artifact of a single annotator's scoring style.

The weakening of the correlation from A0 to GPT-4 is consistent with GPT-4's inflation of A_3: by scoring integration higher on low-integration systems, GPT-4 compresses the A_3 range and reduces the apparent correlation with A_4.

3.5 Gradient primacy (H4)

Mean inter-annotator variance on gradients (Δ_{23}, Δ_{45}): 0.0147. Mean inter-annotator variance on axes (\$A_1\$–A_5): 0.0102. Ratio: 1.44.

H4 is rejected. Gradients are 44\

The failure of H4 is explained by the non-uniform bias structure: annotator biases are axis-dependent (GPT-4 inflates A_3 and A_5 more than A_1 and A_4), so the bias does not cancel in the gradient difference. This means that gradient primacy cannot be demonstrated by variance comparison alone. However, it can be demonstrated by rank correlation: Δ_{45} has higher inter-annotator rank correlation ($\rho = 0.88$) than any individual axis (highest individual axis correlation: A_2 at $\rho = 0.85$). The gradients carry more structurally preserved information than the axes, even if their absolute variance is higher.

3.6 Relational structure preservation

Table 5: Correlation of pairwise distance matrices between annotators.

Pair	Distance matrix ρ
A0–Gemini v2	0.757
A0–GPT-4	0.625
Gemini–GPT-4	0.525

The inter-system distance matrices are positively correlated for all pairs. This means that if two systems are close in A0's scoring, they tend to be close in Gemini's and GPT-4's scoring as well. The neighborhood structure of the organizational space is robust across annotators, even when individual axis scores diverge by up to 0.30.

This result is the most direct evidence for the claim that the organizational space captures genuine relational structure: what is preserved across annotators is not the position of each system, but the pattern of proximities and separations between systems.

4 Discussion

4.1 Relational structure over absolute values

The central finding of this study is that the morphodynamic index produces a relational geometry that is partially preserved across independent annotators. The evidence is threefold: (a) inter-system distance matrices are correlated at $\rho = 0.63$–0.76, meaning that the pattern of proximities and separations between systems is robust even when absolute scores diverge by up to 0.30; (b) the gradient Δ_{45} preserves its rank ordering across all annotator pairs ($\rho = 0.69$–0.88); and (c) the correlation A_3–A_4 persists across all three annotators, suggesting a structural constraint surface in the space.

This result is consistent with the thesis stated in the introduction: what the instrument captures is not a set of metric positions, but an ordinal relational structure. Two systems that are neighbors for one annotator tend to be neighbors for the others. A system that is rigid ($\Delta_{45} \gg 0$) for one annotator is rigid for the others. This is precisely what is meant by a relational invariant: the ordering is preserved under perturbation of the measurement process.

4.2 Status of the two main gradients

The two gradients have different empirical status, and this difference is itself informative.

Δ_{45} (normativity minus revision) behaves as a robust relational invariant. Its rank correlation exceeds 0.70 for all annotator pairs involving A0. It separates the same systems into the same groups regardless of annotator: rigid systems (thermostat, Prussian army, pipeline, bureaucracy) cluster at high Δ_{45}; fluid systems (startup, Occupy, artist collective) cluster at low or negative Δ_{45}. The distinction between "a system that regulates without revising" and "a system that revises more than it regulates" appears to be stable across different scoring agents.

Δ_{23} (propagation minus integration) is robust between A0 and Gemini ($\rho = 0.78$) but fragile with GPT-4 ($\rho = 0.58$). The fragility is traceable to a specific calibration issue: GPT-4 inflates A_3 (integration) on systems where integration is partial or contested, scoring theoretical coordination capacity rather than effective coordination. This is a protocol-dependent sensitivity, not necessarily a property of the gradient itself. The reinforced prompt protocol partially corrects this issue for Gemini (raising $\rho(\Delta_{23})$ from 0.57 to 0.78), suggesting that Δ_{23} could become more robust with better calibration anchors for A_3.

The honest assessment is therefore: Δ_{45} is a relational invariant of the instrument in its current form; Δ_{23} is a candidate invariant whose stability depends on the calibration of the integration axis. Future work should test whether providing explicit anchor examples for A_3 stabilizes Δ_{23} across annotator types.

4.3 Errors as signal

The three error types identified (historical/operational on A_5, endogenous/imposed on A_4, structure/functioning on A_3/A_4) are not noise. They reveal that the instrument's reliability depends not only on the scoring rubric, but on the cognitive frame the annotator brings to the system. An annotator who reads "revision capacity" as "what the system has revised in its history" will score the Prussian army and the Constitution at $A_5 = 1.00$; an annotator who reads it as "what the system can revise during active functioning" will score them at $A_5 = 0.40$–0.60. The reinforced prompt protocol corrects this by making the intended frame explicit. The fact that framing corrections reduce severe divergences by more than half (from 14.5\

4.4 Limitations and the annotator problem

The most significant limitation is the composition of the annotator pool: all three configurations involve LLMs, and two of them (A0 and Gemini) were prompted with comparable instructions. A skeptical reader could argue that the observed convergence reflects shared statistical biases of language models rather than genuine structural properties of the systems. We cannot rule this out with the current data.

However, three observations mitigate this concern. First, the reinforced prompt protocol shows that LLM annotations are sensitive to framing instructions in predictable ways—the same LLM (Gemini) produces markedly different scores before and after a single-paragraph prompt correction. This is inconsistent with a model that simply reproduces a fixed statistical prior. Second, the divergences between Gemini and GPT-4 are substantial (14.4\

A definitive resolution would require fully independent human annotators—ideally domain experts scoring systems within their specialization. This remains as future work.

4.5 Implications for the instrument

The results suggest three modifications for future scoring protocols:

1. **Calibration anchors.** Providing 2–3 scored examples in the prompt (e.g., "the thermostat has $A_3 \approx 0.80$; the supply chain has $A_3 \approx 0.40$") would reduce the calibration divergence on A_3 that destabilizes Δ_{23}.
2. **Mandatory CRITICAL INSTRUCTIONS for ambiguous systems.** The three error types identified (historical/operational, endogenous/imposed, structure/functioning) should be flagged systematically in the scoring protocol, not only post-hoc.
3. **Reformulated H4.** Instead of comparing variances, test whether gradient rank correlations exceed individual axis rank correlations. The current data supports this reformulation for Δ_{45}.

4.6 Limitations

The study uses three LLM-based annotators (including the A0 human–LLM pair). No fully independent human annotators were tested. The corpus ($N = 29$) permits exploratory analysis but not formal statistical inference on family-level properties. The reinforced prompt protocol was applied only to Gemini; GPT-4 results reflect the standard prompt only. Finally, the scoring protocol uses qualitative assessment from textual sources, not quantitative measurement from formal data.

4.7 Conclusion

The gradient-based organizational classification produces a relational geometry—a pattern of orderings, proximities, and separations between systems—that is partially invariant under change of annotator. The normative–revision gradient Δ_{45} is a robust relational invariant; the propagation–integration gradient Δ_{23} is a candidate invariant whose stability depends on the calibration of A_3. The neighborhood structure of the organizational space, the A_3–A_4 constraint surface, and the four morphodynamic families are all preserved across annotators.

These results establish that the instrument measures something structurally consistent: not that the gradients capture fundamental properties of complex systems, but that they produce stable relational coordinates under perturbation of the measurement process. This is the minimal condition for an instrument to be useful as a coordinate system for further inquiry.

The classifier, the scoring protocol, the complete dataset (29 systems × 3 annotators), and all scoring notes with sub-criterion justifications are available at `https://github.com/MMli97/3-index-morphodynami`

5 References

[1] Maassouli, M. (2026). A Gradient-Based Organizational Classification of Complex Systems. *arXiv preprint*. (Paper 1: framework and instrument.)

[2] Maassouli, M. (2026). Empirical Geometry of Organizational Systems: A Gradient-Based Analysis of 28 Heterogeneous Complex Systems. *arXiv preprint*. (Paper 2: empirical study.)